Stock Market Fundamentals Index

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Abstract

1 Introduction

This paper is devoted to study the changes appearing in assets' prices and the relations that could possibly exist between them and their associated fundamental features. Taking into account the information pertaining to assets' fundamentals, is a key step to potentially comprise the set of all known information about assets which is to be publicly accessible. Such set of all known information would be comprised of assets' fundamentals information augmented by the introduced newschapter-based uncertainty information as well as the assets' historical and current prices. Assuming that these three sources of information are the ones accessible to public, in turn enables us to design a set of statistical hypothesis tests to examine the efficient market hypothesis in all its different modes. The importance of our work on efficient market hypothesis is that it provides a guideline for traders and portfolio managers who aim at allocating their investments at optimal positions. Were it to be shown that the market is inefficient, whether on daily, monthly or quarterly frequencies, then this announces the opportunity for the investors to take advantage of the historical and current prices of assets to discover patterns therein which can provide them with a forecast of the future values of the assets. On the contrary, if the market is shown to be efficient, other set of investment strategies should potentially be employed which do not rely on prediction-based approaches of asset prices.

1.1 Notations and Preliminaries

Throughout this paper, scalar quantities, vectors, and matrices are denoted by lowercase lightface (e.g., x), lowercase boldface (e.g., $\mathbf{x}$), and uppercase boldface (e.g., $\mathbf{X}$) letters,

respectively. All vectors are assumed to be column vectors unless stated otherwise. Boldface calligraphic letters (e.g., $\boldsymbol{\mathcal{X}}$) are reserved for tensors, while their lightface calligraphic counterparts (e.g., $\mathcal{X}$) typically denote induced sample spaces.

Tensors are algebraic structures that generalize matrices to higher dimensions. The number of dimensions of a tensor is referred to as its *order* or *mode*. For instance, $\boldsymbol{\mathcal{X}} \in \mathbb{R}^{I_1 \times I_2 \cdots \times I_N}$ denotes an Nth order tensor with $K = \prod_{n=1}^{N} I_n$ elements in total. Analogous to matrix rows and columns, the *mode-n fibers* of a tensor are obtained by fixing all indices except the nth one. Fixing all but two indices yields the *slices* of the tensor.

The Frobenius norm of a tensor $\boldsymbol{\mathcal{X}}$ is defined by

$$\|\boldsymbol{\mathcal{X}}\|_F = \sqrt{\langle \boldsymbol{\mathcal{X}}, \boldsymbol{\mathcal{X}} \rangle} = \sqrt{\sum_{i_1=1}^{I_1} \sum_{i_2=1}^{I_2} \cdots \sum_{i_N=1}^{I_N} x_{i_1 i_2 \ldots i_N}^2}. \tag{1}$$

To reduce tensor order and facilitate matrix computations, one often resorts to *unfolding* (also called *matricization*). The vector unfolding of $\boldsymbol{\mathcal{X}}$, denoted by

$$\mathbf{x} = \texttt{vec}(\boldsymbol{\mathcal{X}}) \in \mathbb{R}^K, \tag{2}$$

reshapes the tensor into a vector. The mode-n matricization of $\boldsymbol{\mathcal{X}}$, given by

$$\boldsymbol{\mathcal{X}}_{(n)} = [\mathbf{f}_1^{(n)}, \mathbf{f}_2^{(n)}, \ldots, \mathbf{f}_{K/I_n}^{(n)}], \tag{3}$$

stacks the mode-n fibers as columns of a matrix of size $I_n \times (K/I_n)$.

The *mode-n product* of a tensor $\boldsymbol{\mathcal{X}} \in \mathbb{R}^{I_1 \times \cdots \times I_N}$ with a matrix $\mathbf{U} \in \mathbb{R}^{J \times I_n}$ is denoted by $\boldsymbol{\mathcal{X}} \times_n \mathbf{U}$ and defined element-wise as

$$(\boldsymbol{\mathcal{X}} \times_n \mathbf{U})_{i_1 \ldots i_{n-1} j i_{n+1} \ldots i_N} = \sum_{i_n=1}^{I_n} x_{i_1 \ldots i_N} u_{j i_n}, \tag{4}$$

yielding a tensor of dimensions $I_1 \times \cdots \times I_{n-1} \times J \times I_{n+1} \cdots \times I_N$. This operation is equivalent to multiplying $\mathbf{U}$ from the left on each mode-n fiber as follows:

$$\boldsymbol{\mathcal{Y}} = \boldsymbol{\mathcal{X}} \times_n \mathbf{U} \quad \Longleftrightarrow \quad \boldsymbol{\mathcal{Y}}_{(n)} = \mathbf{U} \boldsymbol{\mathcal{X}}_{(n)}. \tag{5}$$

When dealing with third-order tensors $\boldsymbol{\mathcal{X}} \in \mathbb{R}^{I_1 \times I_2 \times I_3}$, a useful perspective is to treat them as stacks of frontal slices $\boldsymbol{\mathcal{X}}^{(n)} \in \mathbb{R}^{I_1 \times I_2}$ for $n = 1, 2, \ldots, I_3$. Define the operator

$$\texttt{unfold}(\boldsymbol{\mathcal{X}}) := [\boldsymbol{\mathcal{X}}^{(1)}, \boldsymbol{\mathcal{X}}^{(2)}, \ldots, \boldsymbol{\mathcal{X}}^{(I_3)}]^\top, \tag{6}$$

which stacks all frontal slices vertically. The corresponding inverse operator `fold` satisfies

$$\texttt{fold}(\texttt{unfold}(\boldsymbol{\mathcal{X}})) := \boldsymbol{\mathcal{X}}. \tag{7}$$

Two widely used tensor decompositions are the Tucker and the polyadic (or CP) decompositions. The Tucker decomposition approximates a third-order tensor $\boldsymbol{\mathcal{X}}$ by a dense core

tensor $\boldsymbol{\mathcal{H}} \in \mathbb{R}^{R_1 \times R_2 \times R_3}$ and factor matrices $\mathbf{A} \in \mathbb{R}^{I_1 \times R_1}$, $\mathbf{B} \in \mathbb{R}^{I_2 \times R_2}$, and $\mathbf{C} \in \mathbb{R}^{I_3 \times R_3}$ as follows:

$$\boldsymbol{\mathcal{X}} \approx \widehat{\boldsymbol{\mathcal{X}}} = \boldsymbol{\mathcal{H}} \times_1 \mathbf{A} \times_2 \mathbf{B} \times_3 \mathbf{C}. \tag{8}$$

Alternatively, the polyadic decomposition expresses the tensor as a sum of rank-one components,

$$\boldsymbol{\mathcal{X}} \approx \widehat{\boldsymbol{\mathcal{X}}} = \sum_{r=1}^{R} \mathbf{a}_r \circ \mathbf{b}_r \circ \mathbf{c}_r, \tag{9}$$

where the symbol $\circ$ denotes the vector outer product and R is the rank of the approximation. When R is the minimal number of components required to achieve equality, the decomposition is referred to as the canonical polyadic decomposition (CPD).

Finally, a third-order tensor $\boldsymbol{\mathcal{X}}$ is said to be rank-one if it admits the representation

$$\boldsymbol{\mathcal{X}} = \mathbf{a} \circ \mathbf{b} \circ \mathbf{c}, \tag{10}$$

for some vectors $\mathbf{a} \in \mathbb{R}^{I_1}$, $\mathbf{b} \in \mathbb{R}^{I_2}$, and $\mathbf{c} \in \mathbb{R}^{I_3}$.

2 Encoding Fundamentals Information

In this section, we develop a methodology for embedding stock market fundamentals into structured representations and identifying a subset of features that preserves the latent economic structure of the market. The ultimate objective is to facilitate the construction of a volatility index grounded in fundamental information. The selected features are required to align with a reference taxonomy of economic activity, namely the Global Industry Classification Standard (GICS), thus providing a bridge between qualitative categorization and quantitative representation.

2.1 Feature Selection via Clustering

Let $S = \{S_1, S_2, \ldots, S_N\}$ denote the collection of assets observed in the market, and let $F = \{f_1, f_2, \ldots\}$ denote the universe of available fundamental features. For each asset S_i, let $\mathbf{d}_i \in \mathbb{R}^{|F'|}$ denote the vector of observed values for a subset $F' \subset F$. These vectors are assembled into a design matrix $\mathbf{D} \in \mathbb{R}^{N \times |F'|}$ defined as

$$\mathbf{D} = \begin{bmatrix} \mathbf{d}_1^\intercal \\ \mathbf{d}_2^\intercal \\ \vdots \\ \mathbf{d}_N^\intercal \end{bmatrix} = \begin{array}{c} \\ S_1 \\ S_2 \\ \vdots \\ S_N \end{array} \begin{array}{c} \begin{array}{cccc} f_1^\star & f_2^\star & \cdots & f'_{|F'|} \end{array} \\ \begin{bmatrix} d_{11} & d_{12} & \ldots & d_{1|F'|} \\ d_{21} & d_{22} & \ldots & d_{2|F'|} \\ \vdots & \vdots & \ddots & \vdots \\ d_{N1} & d_{N2} & \ldots & d_{N|F'|} \end{bmatrix} \end{array}.$$

The goal is to determine a subset $F^\star \subset F$ such that the clustering structure inferred from $\mathbf{D}$ closely matches the reference structure given by GICS sub-industry classification. Formally, let $\mathcal{C}(F')$ denote the number of clusters inferred from $\mathbf{D}$ via a clustering algorithm, and let $\mathcal{G}$ denote the cardinality of the GICS sub-industry partition (i.e., $\mathcal{G} = 158$). The

optimal subset is defined by

$$F^\star = \arg\min_{F' \subset F} |\mathcal{C}(F') - \mathcal{G}| \,. \tag{11}$$

This defines our *best-match criterion*, which balances parsimony in feature selection with fidelity to external economic structure. A common alternative to this discrete-matching approach is to measure partition similarity using adjusted Rand index or normalized mutual information, though such refinements are omitted here in favor of interpretability.

Traditional clustering algorithms such as k-means require the number of clusters to be specified *a priori*, which is incompatible with our goal of estimating structure from the data itself. To address this, we employ the *affinity propagation* algorithm [**?**], which treats every observation as a potential cluster exemplar and seeks to minimize an energy function through iterative message passing. This framework automatically determines the number of clusters, with control exerted through a tunable preference parameter. Technical details and convergence analysis are provided in [**?**].

An additional challenge lies in the representation of temporal information. The matrix $\mathbf{D}$ discards the temporal dynamics of each fundamental feature by aggregating values into a single statistic per asset. While this introduces a form of statistical coarsening, it is justified by the relative stationarity of taxonomic classification over short horizons. Sector and sub-industry affiliations typically undergo redefinition on a 4-5 year timescale, often in response to shifts in macroeconomic conditions or technological innovation. To reflect this, we construct $\mathbf{D}$ by averaging quarterly values of each feature from the first quarter of 2016 to the fourth quarter of 2019.[1]

For each candidate subset $F' \subset F$, the design matrix $\mathbf{D}$ is constructed and subjected to affinity propagation clustering. The resulting number of clusters is then compared to $\mathcal{G} = 158$. The subset $F^\star$ is selected according to the best-match criterion. While dimensionality reduction methods such as PCA, Lasso, or autoencoders could be considered, we intentionally avoid these to preserve economic interpretability of the features. Moreover, robustness checks revealed that $F^\star$ remained stable across perturbed data subsets, indicating selection consistency.

In total, over 600 distinct quarterly and annual features were considered. Through extensive search and evaluation, a 40-feature subset $F^\star$ was identified, as detailed in Table **??**. The affinity propagation algorithm applied to $\mathbf{D}$ constructed from $F^\star$ yielded 156 clusters, just two fewer than the GICS-defined sub-industry count. This near match supports the claim that $F^\star$ provides a structurally faithful representation of economic heterogeneity across the market.

2.2 Market Fundamentals Index

We now introduce a tensor-based volatility index that summarizes the aggregate dynamics of stock market fundamentals over time. This *Market Fundamentals Index* (MFI) parallels the FCIXX developed in earlier, but it is derived from fundamental (accounting-based) data rather than asset returns.

[1] Fundamental data retrieved from Compustat via the WRDS platform.

Table 1: Selected fundamental features used for constructing the design matrix.

Feature	Feature
Acquisitions	Inventories - Total
Assets - Other - Total	Investing Activities - Net Cash Flow
Assets and Liabilities	Investing Activities - Other
Capital Expenditures	Long-Term Debt - Total
Cash and Short-Term Investments	Long-Term Debt - Issuance
Cash Dividends	Liabilities Netting Other Adjustments
Common/Ordinary Equity - Total	Noncontrolling Interest
Comprehensive Income - Total	Non-Operating Income (Expense) - Total
Cost of Goods Sold	Operating Activities - Net Cash Flow
Debt in Current Liabilities	Operating Income Before Depreciation
Earnings Per Share (Basic)	Preferred/Preference Stock (Capital) - Total
Earnings Per Share (Diluted)	Pretax Income
Excess Tax Benefit of Stock Options	Receivables - Total
Extraordinary Items	Sale of Common and Preferred Stock
Financing Activities - Net Cash Flow	Sale of Investments
Funds from Operations - Other	Sale of PPE and Investments - Gain/Loss
Quarterly Income Before Extraordinary Items	Sales/Turnover (Net)
Annual Income Before Extraordinary Items	Short-Term Investments - Total
Income Taxes	Special Items
Intangible Assets - Total	Stockholders Equity

Fig. 1: Schematic of a third-order market fundamentals design tensor.

Let $\boldsymbol{\mathcal{D}} \in \mathbb{R}^{N \times |F^\star| \times |\mathcal{T}|}$ denote a third-order design tensor storing the temporal trajectories of selected fundamental features for N assets, where $|\mathcal{T}|$ is the number of time periods (e.g., $|\mathcal{T}| = 120$ quarters). The structure of this tensor is illustrated in Figure **??**.

To characterize the structural complexity of $\boldsymbol{\mathcal{D}}$ and inform the construction of MFI, we begin by applying a polyadic decomposition as follows:

$$\boldsymbol{\mathcal{D}} \approx \widehat{\boldsymbol{\mathcal{D}}} = \sum_{r=1}^{R} \mathbf{s}_r \circ \mathbf{f}_r \circ \mathbf{t}_r, \tag{12}$$

where $\mathbf{s}_r \in \mathbb{R}^N$, $\mathbf{f}_r \in \mathbb{R}^{|F^\star|}$, and $\mathbf{t}_r \in \mathbb{R}^{|\mathcal{T}|}$ denote the rth rank-one component. As shown in Figure **??**, the relative error of approximation remains significantly high even for $R = 100$, indicating that $\boldsymbol{\mathcal{D}}$ is not well approximated by low-rank additive components. This suggests that the interactions among stocks, fundamentals, and time exhibit non-separable and highly nonlinear structure.

Motivated by this observation, we instead employ Tucker decomposition, which generalizes matrix SVD to higher-order tensors by introducing a core tensor to capture multiway interactions. Specifically, we write

$$\boldsymbol{\mathcal{D}} \approx \widehat{\boldsymbol{\mathcal{D}}} = \boldsymbol{\mathcal{H}} \times_1 \mathbf{S} \times_2 \mathbf{F} \times_3 \mathbf{T}, \tag{13}$$

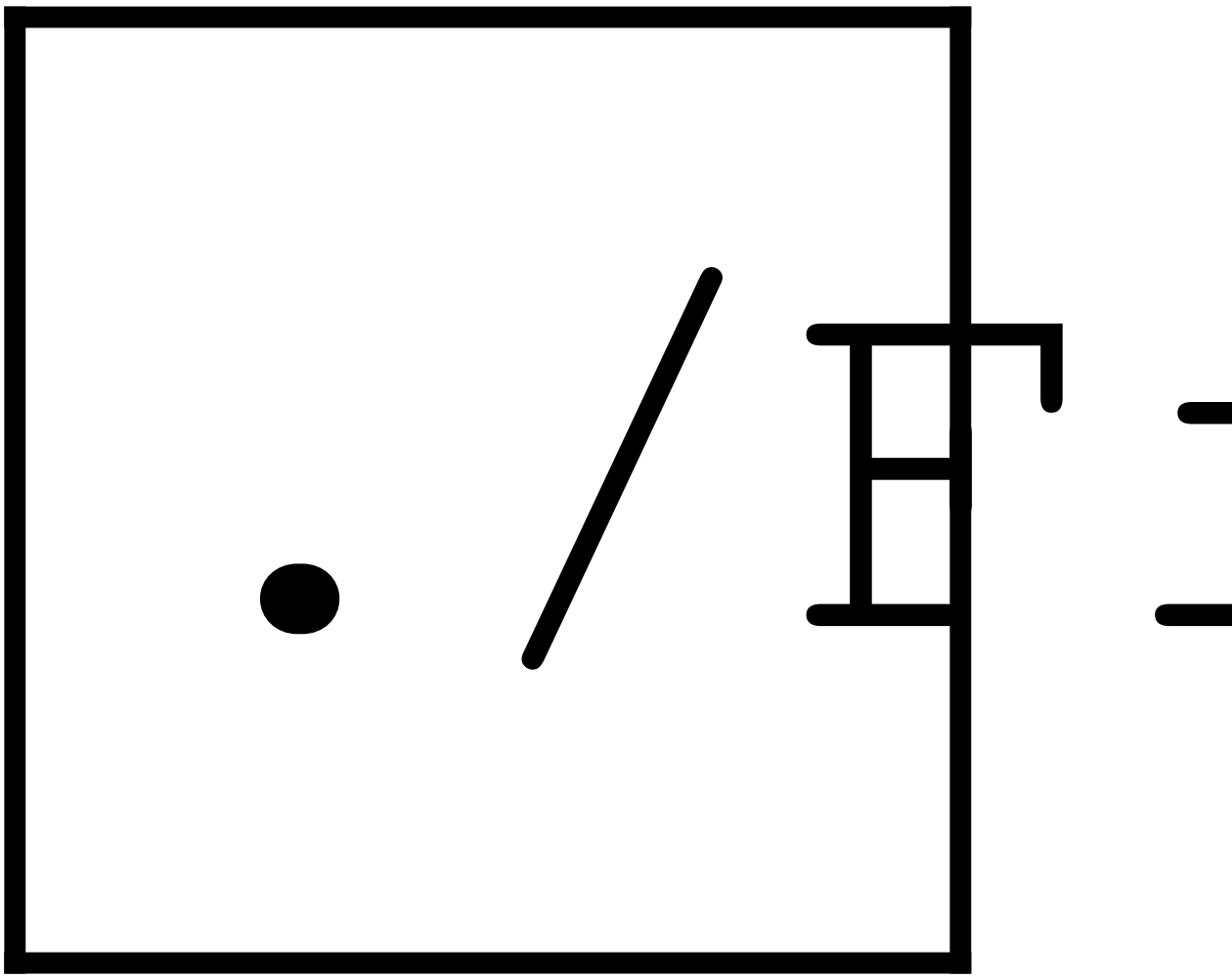

Fig. 2: Plot of the relative error of polyadic decomposition of $\boldsymbol{\mathcal{D}}$ for different values of R.

where $\boldsymbol{\mathcal{H}} \in \mathbb{R}^{R_1 \times R_2 \times R_3}$ is the core tensor, and $\mathbf{S} \in \mathbb{R}^{N \times R_1}$, $\mathbf{F} \in \mathbb{R}^{|F^\star| \times R_2}$, and $\mathbf{T} \in \mathbb{R}^{|\mathcal{T}| \times R_3}$ are orthonormal factor matrices. The multilinear rank vector is defined as $\mathbf{R} = [R_1, R_2, R_3]^\intercal$. The Tucker decomposition is obtained by solving the optimization problem

$$\min_{\boldsymbol{\mathcal{H}}, \mathbf{S}, \mathbf{F}, \mathbf{T}} \left\| \boldsymbol{\mathcal{D}} - \boldsymbol{\mathcal{H}} \times_1 \mathbf{S} \times_2 \mathbf{F} \times_3 \mathbf{T} \right\|_F . \tag{14}$$

To preserve economic interpretability, we fix $R_1 = R_2 = 69$, matching the number of industry categories in the GICS taxonomy. This alignment is intended to capture intra-industry correlations across assets and features. We then vary R_3 and evaluate the relative

approximation error defined by

$$\mathrm{rel}_{\mathrm{error}} = \frac{\left\| \boldsymbol{\mathcal{D}} - \widehat{\boldsymbol{\mathcal{D}}} \right\|_F}{\left\| \boldsymbol{\mathcal{D}} \right\|_F}. \tag{15}$$

Rather than minimizing this error to the lowest achievable level, we intentionally select a moderate value of R_3 that filters out noise and captures only the most persistent components. Based on our empirical evaluation, we select $R_3 = 20$, which yields a relative approximation error of 6.8%.

The temporal factor matrix $\mathbf{T}$ now contains R_3 orthonormal time series of length $|\mathcal{T}|$. Each column captures a distinct latent mode of variation in the temporal dimension of the tensor. To aggregate these into a single index, we define the Market Fundamentals Index by taking the average magnitude across the R_3 components at each time t, whose formal definition is provided below.

Definition 1 (MFI)**.** *Let $\boldsymbol{\mathcal{D}}$ be a third-order tensor of market fundamentals smoothed via Tucker decomposition as $\widehat{\boldsymbol{\mathcal{D}}} = \boldsymbol{\mathcal{H}} \times_1 \mathbf{S} \times_2 \mathbf{F} \times_3 \mathbf{T}$, with $\mathbf{T} \in \mathbb{R}^{|\mathcal{T}| \times R_3}$. The Market Fundamentals Index (MFI) is defined by*

$$\mathrm{MFI}(t) := \frac{1}{R_3} \sum_{k=1}^{R_3} \left| \mathbf{T}_{tk} \right|, \tag{16}$$

for $t = 1, \ldots, |\mathcal{T}|$.

The absolute value ensures that phase shifts or sign reversals do not cancel signal strength, and the averaging operation yields a smoothed, aggregate representation of the total volatility embedded in the fundamentals. As in the case of FCIX_t, we denote the time series realization of this statistic by MFI_t.

Figure **??** displays the quarterly realizations of MFI from January 1990 through December 2019. The time series reveals distinct structural episodes, marked by substantial shifts in the volatility of firm-level fundamentals. Notably, the index remains relatively stable from 1990 through the early 2000s, followed by a sharp regime break in 2002-2003, after which a sustained increase in MFI levels is observed. A pronounced spike appears during the 2007-2009 financial crisis, peaking precisely at the apex of the 2008 global meltdown. The elevated levels in this period reflect heightened dispersion and instability in accounting fundamentals, consistent with widespread balance sheet stress across sectors.

Post-crisis, the MFI declines steadily, stabilizing around 2013, where it enters a low-volatility regime with relatively minor fluctuations. A modest upward drift is again visible beginning in 2018, potentially foreshadowing structural shifts leading into the 2020 pandemic period. These patterns suggest that MFI effectively captures systemic dislocations in fundamental data and exhibits temporal alignment with major financial crises. The dynamics of this index and its potential predictive content are further explored in subsequent empirical tests.

Fig. 3: Plot of the quarterly MFI over the period January 1990-December 2019.

3 Statistical Dependence Between MFI and FCIX

We now examine the empirical relationship between the MFI and the FCIX, with the goal of determining whether their co-movement reflects genuine statistical dependence. Identifying such dependence would support the hypothesis that fluctuations in market fundamentals are structurally linked to volatility in asset prices, thereby providing empirical grounding for the efficient market hypothesis explored in the following sections.

We begin by computing the sample cross-correlation between the quarterly time series $\{\mathrm{MFI}_t\}$ and $\{\mathrm{FCIX}_t\}$ over a range of lags. Figure **??** illustrates the resulting correlation coefficients from -10 to $+10$ quarters. A strong and symmetric structure is observed, with peak correlation of approximately 0.75 at lag one, indicating near-contemporaneous alignment and persistence of shared variation over time.

While correlation provides a preliminary indication of synchrony, it is limited to capturing linear dependence and cannot rule out the possibility of spurious alignment. To test for general dependence, including nonlinear and higher-order structures, we employ a nonparametric framework based on empirical distribution comparisons. Specifically, we adopt the consistent testing methodology proposed in [**? ?**], which compares the joint empirical distribution with the product of its marginals using two divergence-based test statistics.

Let $\{(\mathrm{FCIX}_t, \mathrm{MFI}_t)\}_{t=1}^n$ be a sequence of i.i.d. observations drawn from an unknown joint distribution $\mathbb{G}_{\mathrm{FCIX,MFI}}$. Denote the corresponding marginal distributions by $\mathbb{G}_{\mathrm{FCIX}}$ and $\mathbb{G}_{\mathrm{MFI}}$. The null hypothesis of independence is expressed as:

$$\mathrm{H}_0^{\mathtt{INDP}}: \quad \mathbb{G}_{\mathrm{FCIX,MFI}} = \mathbb{G}_{\mathrm{FCIX}} \times \mathbb{G}_{\mathrm{MFI}}.$$

Let $\mathcal{P}_n = \{A_{n,1}, \ldots, A_{n,m_n}\}$ and $\mathcal{Q}_n = \{B_{n,1}, \ldots, B_{n,m'_n}\}$ denote measurable partitions of $\mathbb{R}_{\geq 0}$, constructed to discretize the support of the empirical distributions. The first test statistic is based on the L_1 distance between the joint and product measures:

$$L_n := \sum_{A \in \mathcal{P}_n} \sum_{B \in \mathcal{Q}_n} \left| \mathbb{G}_{n,\mathrm{FCIX,MFI}}(A \times B) - \mathbb{G}_{n,\mathrm{FCIX}}(A)\, \mathbb{G}_{n,\mathrm{MFI}}(B) \right|. \tag{17}$$

The empirical probabilities are defined by:

$$\mathbb{G}_{n,\mathrm{FCIX,MFI}}(A \times B) := \frac{1}{n} \sum_{t=1}^{n} \mathbb{I}[(\mathrm{FCIX}_t, \mathrm{MFI}_t) \in A \times B], \tag{18}$$

$$\mathbb{G}_{n,\mathrm{FCIX}}(A) := \frac{1}{n} \sum_{t=1}^{n} \mathbb{I}[\mathrm{FCIX}_t \in A], \tag{19}$$

$$\mathbb{G}_{n,\mathrm{MFI}}(B) := \frac{1}{n} \sum_{t=1}^{n} \mathbb{I}[\mathrm{MFI}_t \in B]. \tag{20}$$

./Figures/Fig_Cross_Corr_Quarters.pdf

Fig. 4: Sample cross-correlation between quarterly FCIX_t and MFI_t over the period January 1990?December 2019.

As an alternative, we compute a log-likelihood divergence statistic, which measures the Kullback?Leibler distance between the empirical joint and product-marginal distributions:

$$I_n := \sum_{A \in \mathcal{P}_n} \sum_{B \in \mathcal{Q}_n} \mathbb{G}_{n,\mathrm{FCIX},\mathrm{MFI}}(A \times B) \log \left(\frac{\mathbb{G}_{n,\mathrm{FCIX},\mathrm{MFI}}(A \times B)}{\mathbb{G}_{n,\mathrm{FCIX}}(A)\, \mathbb{G}_{n,\mathrm{MFI}}(B)} \right). \tag{21}$$

Under the null hypothesis of independence, both L_n and I_n converge in distribution, and critical values can be approximated either via permutation or using asymptotic theory.

Table **??** reports the results of both tests applied to $\{\mathrm{FCIX}_t\}_{t=1}^{120}$ and $\{\mathrm{MFI}_t\}_{t=1}^{120}$, using $m_n = m'_n = 6$ discretization bins per dimension. This choice is informed by the market segmentation identified in Section **??**, where six distinct macroeconomic regimes were found across the three-decade horizon.

Table 2: Results of independence tests between quarterly FCIX_t and MFI_t.

	L_1-based	Log-likelihood
Test statistic	0.839	1.073
Critical thresholds		
1% level	0.514	0.464
5% level	0.492	0.416
10% level	0.479	0.391

Both test statistics exceed their respective critical values at the 1% significance level, leading to the rejection of the null hypothesis $\mathrm{H}_0^{\mathtt{INDP}}$. This provides strong statistical evidence that FCIX_t and MFI_t are dependent, indicating that price-based market volatility and structural shifts in accounting fundamentals are jointly distributed and possibly mutually informative. These findings serve as a critical empirical foundation for evaluating market efficiency in the subsequent section.

4 Predictive Modeling for Fundamentals

5 Conclusion

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